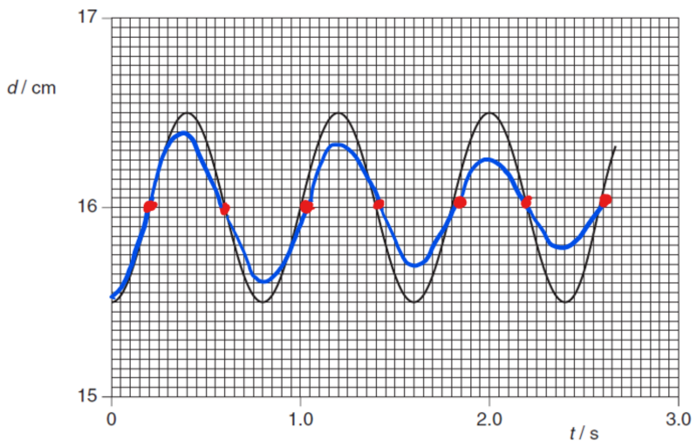
4(a)	amplitude = 0.50 cm	[1]
b(i)	T = 0.8 s $\omega = 2\pi / T$ = 7.85 rad s^{-1} $v = \omega \sqrt{(x_0^2 - x^2)}$ $= 7.85 \times \sqrt{(\{0.5 \times 10^{-2}\}^2 - \{0.2 \times 10^{-2}\}^2)}$ = 3.6 cm s^{-1}	[1] [1 – ecf for (a)] [1 – ecf for (a)]
(ii)	d = 15.8 cm	[1]
(c)	• same period { Accept: small increase in period) • (For same period) : displacement of damped oscillator is always smaller than that on Fig.4.2 (ignore first T/4) • peak decreases (exponentially) with time  	[1] [1] [1]

5	light intensity has maximum value at 0°, 180°, 360° and zero intensity at 90°,	